

stantial specification upgrade.

"The shape of desktop instruments has not changed much over the years. In the development of the instrument of the contemporary era, new-era engineers also hope that the new technology can be applied to professional equipment."

"Latest equipment will be integrated with a variety of functions such as touchscreens, mobile devices, cloud connectivity and intelligent forecasting, to provide more features than at any time in the past and more significant technology advantages," explains Chandmal Goliya, director, Kusam Electrical Industries Ltd.

UI updates

Chivukula says that the current trend is to delink the user interface (UI) from the device. This way, any good that happens to the computer also gets transferred to the equipment in use. If your computer

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has the latest high-performance processor and high-resolution touchscreen, then your instrument inherits it too.

You do not need to know about where the hardware is, so in our case we delinked the UI from the actual instrument and that gives you a lot more convenience. Let us say you are working on a signal, you can move the scope beyond our desk and, by using wireless connectivity, get it to display the data on your desk.

Hardly anyone uses scopes standalone these days. They use it together with a function generator and other equipment. With the modern way of testing, the delinked UI makes operation easier even with multiple products.

Education focus

Touchscreens were already seen in the higher-end equipment but we have now started seeing these in the lower-bandwidth segments as well with equipment like InfiniVision 3000-T series.

"For the education market and hobbyists, our principals are planning to integrate more instruments in one unit, with Android, Wi-Fi support and more apps being planned," explains Goliya.

A product like Analog Discovery developed by Digilent, Analog Devices and Xilinx gives students the option to discover signals from their class itself, without having to get equipment inside the lab. "I feel the notions of separate classes and labs were for the 1990s or so, where we had bulky computers and instruments. But now almost everything is pocketable. Consumer electronic developments in the last ten years have been in terms of technology that we need to leverage for lab equipment to enable in-class teaching for people to work with real signals," says Chivukula.

Oscilloscopes: The last one standing

Ushani adds that oscilloscopes are no longer designed to only display the variation of signals with respect to time. Multiple functions and options are continually being added to newer designs.

In addition, other instruments are becoming integral features of some of today's oscilloscopes. Digital-signal-analysis tools such as logic analysers and pattern generators, arbitrary waveform generators and other test and measurement instruments are among features being offered by several manufacturers.

"This trend is likely to continue until an oscilloscope is all the test equipment that is needed in an electronic laboratory," says Ushani. **EFY**

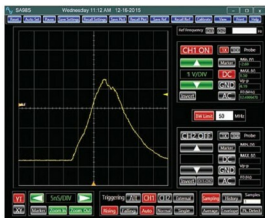


Fig. 4: Display of a 10ns pulse with a 50MHz bandwidth oscilloscope



Fig. 5: A typical spectral and oscilloscope plot of the same signal